Wayfair Rugs Market Intelligence: AI Agent Demo

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Executive Summary

Overview of Market Intelligence Agents

🎯Objective:

Help Wayfair's Rugs category team **automate content trend discovery**, **competitor monitoring**, and **idea generation for social media** content by building AI agents.

🧠My Solution

Built **three** connected AI agents that feed into one Market Intelligence Dashboard.

⚙️ Core Outputs

- ✔ Trend Discovery Agent - Sample Output Link
- ✔ Competitor Monitoring Agent - Sample Output Link
- ✓ AI Insights & Content Agent - Sample Output Link
- ✓ Integrated Dashboard - Sample Output Link

🔍 Key Insights

- Synthetic options like **Polyester** are seeing rapid adoption with a **12.05% growth rate**.
- **Earth-inspired palettes** (sand, terracotta, sage) are **replacing** previously dominant **cool gray tones**
- Both **Amazon and Walmart** focus heavily on "Machine Washable" and "Non-Slip" features. Wayfair does not.

🧠Future Improvements

- Category expansion beyond rugs
- Implementation of SKU analysis to understand the relationship between shifts in trends and shifts in sales

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Agent 1: Consumer Trend Discovery

From Raw Market Signals → Emerging Rug Trends

🎯Objective:

To **identify** both **emerging** and **sustained rug trends** by analyzing product collections and listings.

🧠Input Prompt Example

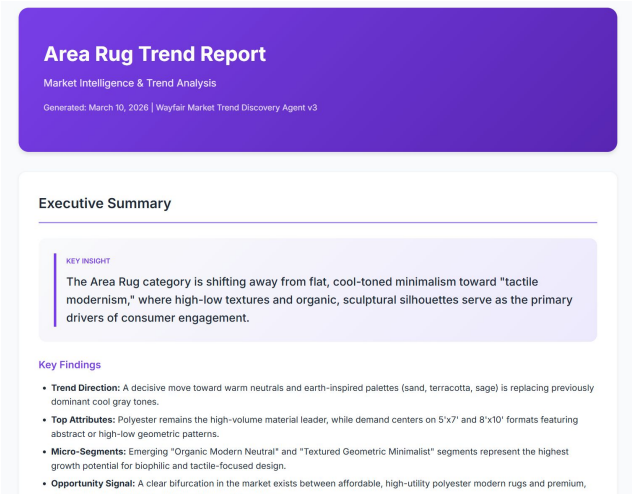
Rug Category (“Area Rug”) + Amazon Collection/Direct Product URLs

⚠ Keep in Mind / Input Notes

Currently optimized **exclusively** for Amazon product/collection links and the Extern API

💡 Improvements / Expansion Opportunities

Integrate direct scraping for **other retailers** to better understand trends in the general public. Integrate direct scraping for **high-end rug boutiques** and **design catalogs** to capture luxury trends early,



Area Rug Trend Report

Market Intelligence & Trend Analysis

Generated: March 10, 2026 | Wayfair Market Trend Discovery Agent v3

Executive Summary

KEY INSIGHT

The Area Rug category is shifting away from flat, cool-toned minimalism toward "tactile modernism," where high-low textures and organic, sculptural silhouettes serve as the primary drivers of consumer engagement.

Key Findings

- **Trend Direction:** A decisive move toward warm neutrals and earth-inspired palettes (sand, terracotta, sage) is replacing previously dominant cool gray tones.
- **Top Attributes:** Polyester remains the high-volume material leader, while demand centers on 5'x7' and 8'x10' formats featuring abstract or high-low geometric patterns.
- **Micro-Segments:** Emerging "Organic Modern Neutral" and "Textured Geometric Minimalist" segments represent the highest growth potential for biophilic and tactile-focused design.
- **Opportunity Signal:** A clear bifurcation in the market exists between affordable, high-utility polyester modern rugs and premium,

📄View JSON Workflow

🖼️ View Sample Output

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Agent 2: Competitor Monitoring

From Competitor Collection/Product Pages → Price & Assortment Gap Analysis

🎯Objective:

To **monitor competitor** rug **launches** and pricing **strategies**, as well as **identify gaps** and **white space opportunities**.

🧠Input Prompt Example

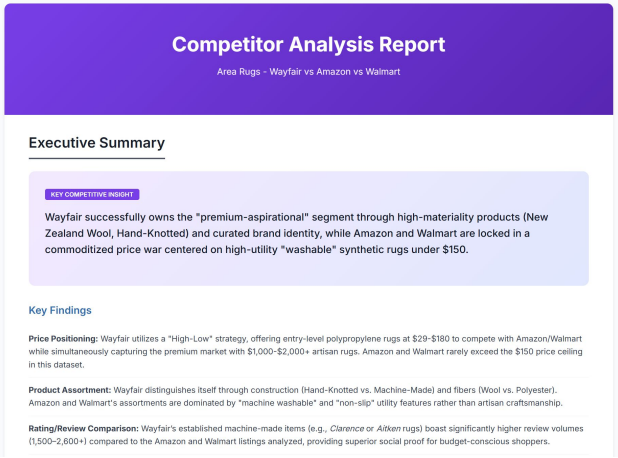
Rug Category (“Area Rug”) + Amazon Collection/Product URLs + Walmart Collection/Product URLs

⚠ Keep in Mind / Input Notes

The agent automatically visits **pre-configured** Wayfair collection URLs based on the chosen category for a 3-way comparison

💡 Improvements / Expansion Opportunities

Connect insights directly to a **pricing tool** to suggest real-time adjustments based on competitor markdowns



Competitor Analysis Report

Area Rugs - Wayfair vs Amazon vs Walmart

Executive Summary

KEY COMPETITIVE INSIGHT

Wayfair successfully owns the "premium-aspirational" segment through high-materiality products (New Zealand Wool, Hand-Knotted) and curated brand identity, while Amazon and Walmart are locked in a commoditized price war centered on high-utility "washable" synthetic rugs under $150.

Key Findings

Price Positioning: Wayfair utilizes a "High-Low" strategy, offering entry-level polypropylene rugs at $29-$180 to compete with Amazon/Walmart while simultaneously capturing the premium market with $1,000-$2,000+ artisan rugs. Amazon and Walmart rarely exceed the $150 price ceiling in this dataset.

Product Assortment: Wayfair distinguishes itself through construction (Hand-Knotted vs. Machine-Made) and fibers (Wool vs. Polyester). Amazon and Walmart's assortments are dominated by "machine washable" and "non-slip" utility features rather than artisan craftsmanship.

Rating/Review Comparison: Wayfair's established machine-made items (e.g., *Clarence* or *Aitken* rugs) boast significantly higher review volumes (1,500-2,600+) compared to the Amazon and Walmart listings analyzed, providing superior social proof for budget-conscious shoppers.

📄View JSON Workflow

🖼️ View Sample Output

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Agent 3: AI Analysis and Content

From Trend Discovery & Competitor Monitoring → Content & Marketing Ideas

🎯Objective:

To **translate** consumer **trends** and competitor **signals** from various retailers into **strategic recommendations and content ideas**

🧠Input Prompt Example

Rug Category ("Area Rug") + Focus Area (Optional, "Sustainable Materials") + Uploads: [Trend Report + Competitor Report]

⚠ Keep in Mind / Input Notes

The agent **requires both** reports in order to provide a comprehensive strategy report

💡 Improvements / Expansion Opportunities

Generate **role-specific** recommendations (merchandising, marketing, etc.)

🎬 Aesthetic Video "Vibe Boards"

AESTHETIC VIDEO
Sunday Morning Alchemy

Soundscape: *The soft crackle of a vinyl record, the rhythmic clink of a coffee cup being set down, and the distant hum of a city waking up. Acoustic guitar strums melt into the sound of rain on a skylight—ASMR meets lo-fi.*

🎬 Visual Beats

1. A close-up of bare feet sinking into a plush, undyed wool rug.,2. Morning light streaming through sheer linen curtains, casting moving shadows on a reclaimed wood console.,3. Hands wrapping around a hand-thrown ceramic mug, steam rising in slow motion.,4. A slow pan across a gallery wall of mismatched frames, each holding a memory (a Polaroid, a child's drawing, a vintage postcard),.5. The camera lingers on a well-loved armchair, its fabric slightly worn at the edges—proof of a thousand lazy afternoons.

Manifesto: "The best homes aren't designed. They're *lived* ✨"

AESTHETIC VIDEO
The Patina of Time

Soundscape: *The sound of a paintbrush gliding over canvas, the gentle scrape of a wooden chair being pulled out, and the ambient noise of a bustling flea market. A hauntingly beautiful cover of 'Dream a Little Dream of Me' plays in the background.*

🎬 Visual Beats

1. A macro shot of a weathered leather armchair, its creases telling a story.,2. Sunlight hitting a brass doorknob, its surface worn smooth by decades of use.,3. A slow zoom on a dining table set with mismatched vintage plates, each with its own history.,4. The camera follows a hand as it traces the grain of a reclaimed wood coffee table.,5. A final shot of a bookshelf, its shelves sagging slightly under the weight of well-loved novels and trinkets.

Manifesto: "Why the most beautiful homes are the ones that show their age ✨"

📄 View JSON Workflow

🖼️ View Sample Output

Market Intelligence Dashboard

Bringing All Agents Together → Live Insights for Wayfair's Rugs Team

🎯Objective:

To provide Wayfair teams a **live dashboard** that **combines trends, competitors,** and **strategic recommendations.**

⚙️ How This Works

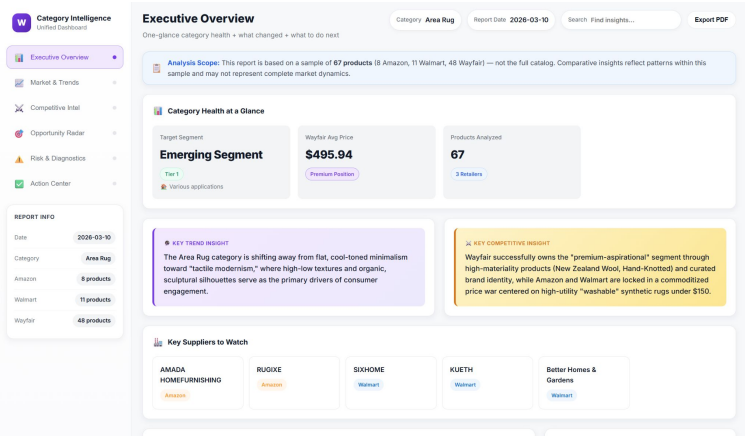
Each AI Agent write structured outputs to a centralized data store. Dashboard auto-updates via connected n8n workflows. Executive Summary Tab automatically aggregates insights from Trend Discovery and Competitor Monitoring Agents

⚠ Keep in Mind / Implementation Notes

Dashboard follows pre-defined template, designed for **quick analysis**

💡 Improvements / Expansion Opportunities

Implement **historical comparisons** to show changes in insights over time.



📄View JSON Workflow

🖼️View Dashboard

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Reflections & Future Improvements

What I Learned and How I Can Make It Better

🧭 Key Learnings

- I learned how to **utilize AI to translate market signals into actionable insights** for business teams.
- I learned firsthand how data **informs** design, assortment, and pricing decisions
- I gained hands-on experience **building** AI agents and dashboards
- I learned how **automation** can be used to **simplify** the task of monitoring products of trends manually

🚀 Future Improvements / Next Steps

- **Integrate** SKU performance metrics to connect consumer trends with Wayfair sales data.
- **Expand** the system to ingest a higher volume of product URLs, social media posts, and design blogs to increase report accuracy and trend reliability
- **Refine** AI prompts for more brand-aligned recommendations
- **Extend** data sources beyond marketplaces to social and influencer data

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Thank You!